

Simulation for Solar Cells

In this section, continuing from last month's article, we discuss Synopsys' Saber simulation tool, and its advantages for design engineers working in solar/PV segment



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Simulation can help one get key insights into the physics of a solar cell operation, thus enabling engineers to explore the complete range of design alternatives. When we look at the module and system levels, behavioural models allow engineers to examine design trade-offs that can

affect system performance. Currently, simulation plays a huge role in the development of different variants of photovoltaic technology, from solar cell design to system performance, and provides an outlook that will help in future advancements in this area.

The first solar cell intended for commercial use was developed in 1954 by researchers at the Bell Telephone Labo-

ratories. Based on a diffused p-n junction fabricated in silicon, this cell was the first version of a solar cell. At that time, silicon was already a prominent semiconductor used in electronic devices, and its selection as the semiconductor for the first generation of solar cells was justified in view of its rapidly advancing manufacturing technology and excellent electronic and optical properties.